

Charles Herman

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Summary

Mechanical engineering student with hands-on experience in rocketry, mechanical design, and technical presentation. Contributed to hybrid propulsion development and recovery systems, and recently assumed the role of Chief Engineer within The University of Melbourne's high-powered rocket team, ARES. Known for practical problem-solving, reliability, and effective collaboration across technical groups. Two-time mechanical engineering intern at ACE and Toyota. Currently working as an engineer on a caravan-moving robot project at Safety Dave.

Education

- **University of Melbourne**, Masters of Mechanical Engineering *Present*
- **University of Melbourne**, Bachelor of Science (Mechanical Engineering Systems) *2026*

Experience

- **Undergraduate Engineer – Safety Dave** *June 20206 – Present*
Mechanical development and physical manufacturing for a caravan-moving robot. Executing 3D print prototyping, hardware troubleshooting, structural torque specing, assembly optimisation and assembly.
- **Engineering Internship – Toyota Motor Corporation Australia** *Nov 2025 – March 2026*
Working within the Product Planning and Development division to support mechanical design and operational efficiency. Responsibilities include assisting with part fitment, CAD modelling using CATIA, conducting field testing, FEA validation and performing data analysis to support design optimisation.
- **Engineering Internship – Ascot Consulting Engineers** *July 2025 – Nov 2025*
Supporting senior engineers on building services projects, working across multiple disciplines, attending client meetings, and developing practical skills in AutoCAD and Revit.
- **ARES Rocketry Team – Chief Engineer** *July 2025 – Present*
Leading the technical development of high-powered experimental rockets, coordinating multidisciplinary teams and overseeing system integration. Responsibilities include ensuring design robustness, meeting competition requirements, and guiding component testing and validation.
- **ARES Rocketry Team – Propulsion/ Recovery Sub-Team Member** *Mar 2024 – July 2025*
Completed PCR and CDR, and fabricated a spin-cast system for paraffin fuel grain manufacturing. Developed MATLAB-based hybrid motor sizing scripts to determine appropriate combustion chamber dimensions. Technical presentations to the Victorian Rocketry Association, and co-authored technical documentation for IREC competition judges.

Full work history and references available upon request.

Skills

- **Software:** Fusion360, CAM, Solidworks, Onshape, Catia, Abaqus
- **Fabrication:** 3-Axis CNC, Soldering, Metalworking, 3D Printing, composite part manufacturing.
- **Skills:** DFM and CNC, GDT, FMEA, FEA, mechanical design, strain gauging, model analysis, technical report writing, team leadership, adaptability.